

Pritesh Gurjar

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EDUCATION

Case Western Reserve University

Master of Science in Computer Science | GPA: 3.76/4.0

Cleveland, OH

Aug 2024 – May 2026

- **Relevant Coursework:** Generative AI, Distributed Systems, High-Performance Computing, Machine Learning, Deep Learning

TECHNICAL SKILLS

ML Frameworks & Tools: PyTorch, TensorFlow, OpenCV, Hugging Face, Scikit-learn, PySpark, AST

GenAI Engineering: RAG Pipelines, Semantic Search, Prompt Engineering, LangGraph, Multi-Agent Systems, Tool Calling

Backend & Data Engineering: Python (FastAPI), Java (Spring Boot), SQL, MongoDB, Vector Embeddings (Milvus/FAISS)

Infrastructure & HPC: Docker, Kubernetes, Azure, Linux/Unix, CI/CD Pipelines, Distributed Systems, Cloud Load Balancers

EXPERIENCE

AI Research Assistant – Agentic AI Systems

University Technology (UTech), CWRU

Oct 2024 – Present

Cleveland, OH

- Architected a complex **multi-agent system** using **LangGraph** to orchestrate specialized LLM agents, establishing autonomous research and scheduling workflows for 500+ faculty.
- Designed and implemented secure **tool-calling architectures** for LLMs to interface with internal APIs, ensuring FERPA data compliance and achieving a 35% increase in research productivity.
- Developed an evaluation framework using **NDCG and MRR metrics** to benchmark retrieval quality, resulting in a 20% improvement in context relevance through iterative re-ranking optimizations.
- Optimized system performance on Linux servers by standardizing reusable AI components and prompt caching, reducing token costs by 30% and improving inference latency.

Software Engineer – Full stack & Data Engineering

NielsenIQ

Jul 2022 – Dec 2024

Pune, India

- Engineered scalable backend solutions using **Java Spring Boot** and **Python**, optimizing RESTful interfaces to support high-throughput client applications with 99.9% uptime.
- Designed and implemented high-performance data pipelines using **PySpark** and **MongoDB** for downstream ML feature stores, processing and ensuring quality for 10M+ records daily.
- Optimized system architecture by deploying microservices on **Docker and Kubernetes**, facilitating horizontal scaling and reducing API latency by 60% through efficient resource management.
- Integrated cloud load balancers and configured web servers (Nginx) to manage high concurrent traffic, demonstrating expertise in large-scale distributed system deployment.

PROJECTS

EduAssist-AI – Full-Stack Multimodal RAG Pipeline | *React, FastAPI, Whisper, CLIP, FAISS*

2024

- Architected a multimodal GenAI application orchestrating **Whisper v3** (audio transcription), **CLIP** (vision encoding), and **T5** (text generation) for processing lecture videos.
- Transformed unstructured video data into structured knowledge graphs, implementing a RAG retrieval system using **FAISS** for semantic search over complex, diverse content types.
- Designed the high-performance backend using **FastAPI**, achieving sub-200ms retrieval latency and 92% accuracy in context delivery by fine-tuning models on educational datasets.

TestPilot Auto – Agentic Code Generation System | *Python, Selenium, LLM, Abstract Syntax Trees (AST)*

2024

- Built an **agentic workflow** leveraging Code Llama that autonomously analyzes web structures and generates functional Selenium test scripts, reducing manual authoring time by 70%.
- Implemented self-correction capabilities by integrating **tool-calling** to execute generated tests in a sandbox, parsing error logs using Abstract Syntax Trees (AST) for iterative refinement.
- Achieved 85% code syntax accuracy by optimizing context management and prompt engineering techniques, significantly lowering test maintenance overhead.

Library AI Search Agent – Enterprise Retrieval System | *RAG, NoSQL, NLP*

2024

- Engineered an intelligent retrieval agent connecting to large catalog databases, processing over 10,000 queries with a custom ranking algorithm prioritizing academic relevance.
- Optimized the **vector search retrieval** pipeline to handle complex natural language queries, decreasing librarian support tickets by 45% and reducing null-result rates.

ACHIEVEMENTS

Published Researcher: Developed Traffic Sign Recognition using Deep Learning techniques (97% accuracy), published in IRJET.

IBM Gold Winner: Recognized for excellence in Security Technologies Research (2020).